Scalar–Tensor Hydrodynamics

Gravity as the Response of a Vacuum Medium:
A Modified Gravity Framework for Vacuum Viscosity and Cyclic Vacuum Cosmology

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Revised Formalism — Version 3.7

Abstract

We propose a hydrodynamic extension of General Relativity in which the vacuum is modeled as a dynamic, high-energy scalar medium described by a field ϕ. Rather than treating the vacuum as a fixed geometric background plus a constant cosmological term, the framework assigns the vacuum its own stress-energy, pressure, and dissipative response. The formulation separates the Einstein geometric tensor from the vacuum viscous stress tensor, introduces a dynamical scalar-field sector, replaces a fixed cosmological constant with a field-dependent pressure contribution, and allows the effective gravitational coupling to depend on the local vacuum state. The framework is intended to investigate whether cosmological expansion, galactic dynamical anomalies, and the transition between localized and delocalized excitations can arise from one underlying medium. The medium is taken as primary. What are conventionally called particles are treated as localized excitations of it, and spacetime geometry, with Einstein's equation, as the effective description of how its state is experienced. Gravity is proposed to be the medium's response to concentrated energy.

This document is a theoretical proposal, not an established physical theory. The equations below define a research program whose consistency, stability, conservation laws, and observational predictions must be demonstrated.

Preface: What This Paper Is For

The premise

There is no empty space.

Everything this paper describes happens inside a single continuous medium, called here the Ocean. The name is a metaphor, and one feature of real oceans must be discarded at once: this Ocean has no edges. It has no shore and no surface. There is no sky above it, no floor beneath it and no air over it. It is not held in a basin, and it does not sit inside a space that could exist without it. There is no outside. The edge of what we can see is a limit of seeing, not an edge of the Ocean. Where things seem empty, the Ocean is there, quiet. Where they seem full, the Ocean is there, gathered. Nothing in this framework exists apart from it, beside it or beyond it.

What we call matter is the Ocean's own energy, gathered and circulating in place. What physics calls a particle is not an object but a pattern the Ocean keeps re-making. When a particle "moves", the place where the pattern is re-made has shifted, the way a wave crosses a stadium while nobody leaves their seat. Light is a ripple in the same medium. A galaxy is a vast region of gathered energy and current. Every pattern exists against, alongside or within others, adding to or taking from them, and every one is ultimately the Ocean's response to what the rest of it has done. For that reason this document uses the word "particle" only when reporting other authors' theories, and marks it as theirs.

This work began with gravity. Einstein's theory describes what gravity does with extraordinary precision: clocks run slow near a mass, space stretches, paths curve. It does not say why concentrated energy should do this. The proposal here is that gravity is the medium's response. Concentrated energy changes the state of the Ocean around it, the Ocean runs slower there, and every pattern in it, light included,

leans toward where it runs slower. On this view Einstein's equations remain the correct description, and the Ocean is offered as the reason they are correct, as the gas laws stayed correct when atoms were found to explain them. Whether the framework can carry that claim is the test it must pass: every pattern must lean by exactly the same amount, and the stretching of space must match the slowing of time exactly as Einstein requires.

The problem

The same premise bears on the largest open problem in cosmology.

Two of the largest numbers in cosmology are placeholders. Roughly a quarter of the contents of the observable universe is labelled *dark matter*: a substance inferred entirely from its gravitational pull on things we can see, and never once detected directly. Most of the remainder is labelled *dark energy*, which is less a substance than a name for an observation — that cosmic expansion is speeding up when everything we understand says it should be slowing down. Together these two labels cover something like ninety-five per cent of what the observable universe contains. Neither has an agreed physical identity.

Standard cosmology handles this competently. It adds an invisible cold matter component, and a constant term to Einstein's equations, and the resulting fit to observation is superb — good enough that the model deserves the confidence it has. But a superb fit obtained by inserting two unexplained quantities is not the same thing as an explanation. It is a description with two holes in it, and the holes have names.

This paper asks whether both holes might have one filling.

A note on where these numbers came from

There is a precedent in this subject for inserting a quantity because the answer is expected rather than because anything has been measured. In 1917 Einstein added a term to his field equations — the cosmological constant, Λ — for no reason beyond his conviction that the universe, as he conceived it, ought to be static, and the inconvenient fact that his equations refused to make it so. No observation supported the term. It was a number chosen to produce an anticipated result. When Hubble demonstrated a little over a decade later that the observable universe is expanding after all, Einstein abandoned it.

It is worth being precise about what happened next, because the common retelling is wrong in a way that matters. Λ returned in 1998, but not as Einstein's unmeasured assumption quietly surviving. It returned because observations of distant supernovae showed cosmic expansion to be accelerating, and Λ was the simplest term capable of describing that. The acceleration is real and measured. What Λ does not do is explain it: no theory predicts the constant's value, and the various attempts to calculate it from first principles fail by a margin discussed below. The term records what the expansion does and is silent on why.

Dark matter has an entirely separate and considerably stronger pedigree. Fritz Zwicky noticed in 1933 that galaxies in the Coma cluster move too fast for the visible mass to hold them together. Vera Rubin's rotation-curve measurements in the 1970s found the same discrepancy inside individual galaxies. Gravitational lensing, the structure of the cosmic microwave background, and the separation of mass from gas in colliding clusters have since confirmed it from independent directions. The evidence is not thin, and this document does not dispute any of it.

The objection is therefore not that these quantities were invented. It is that in both cases a measured discrepancy was given a name, and the name was then promoted to a substance. Λ is a parameter fitted to the expansion history rather than derived from anything. Dark matter is a material known solely by its gravity, which forty years of increasingly sensitive detectors have failed to find by any other means. Each is a placeholder that has been in service long enough to be mistaken for an answer.

If the framework set out here is even approximately correct, nothing observational changes. The rotation curves remain exactly as measured. So do the lensing maps, the microwave background, and the supernova data establishing acceleration. What dissolves is the pair of substances invoked to account for them: they stop being two unknown constituents of the observable universe and become two behaviours of one medium. The measurements survive intact. Only their cause is reassigned.

Where our two best theories disagree

General Relativity and quantum field theory are the most thoroughly confirmed descriptions of nature ever constructed, and they are known to be mutually incompatible. In ordinary practice this causes little trouble, because the two rarely describe the same situation: one governs the very large, the other the very small, and the regimes where both matter at once — the first instants after the big bang, the interior of a black hole — are exactly the regimes we cannot observe.

There is one glaring exception, and it is directly relevant here. The two theories overlap on the question of what the vacuum is, and there they do not disagree by a little. General Relativity treats empty space as genuinely empty: a smooth, featureless backdrop with no properties of its own. Quantum field theory describes the same emptiness as the most active object in physics — fields in their lowest state, never still, producing effects that have been measured in the laboratory for seventy years.

Ask both theories what that activity should weigh, and the answers differ by something between sixty and a hundred and twenty orders of magnitude, depending on where the calculation is cut off. This is not a discrepancy to be tidied away with a correction factor. It is routinely described as the worst quantitative prediction in the history of science, and it has stood unresolved for decades.

The observation that motivates this document is that the sharpest disagreement between our two best theories is a disagreement about whether the vacuum is a thing. One says no. The other says emphatically yes, and has the laboratory results to support it. What follows takes the second answer seriously and asks a further question that neither theory addresses: if the vacuum is a thing, does it also have mechanical properties — density, pressure, flow, resistance to being stirred — and if so, what would we expect to see?

What a medium can do

General Relativity, read as it is usually taught, treats the vacuum as the absence of things: a stage that can be curved by what sits on it but has no density, pressure or motion of its own. A medium does things a stage cannot. It flows. It drags. It carries waves. They can be stirred, and they push back when you stir them. Each of those behaviours is a candidate for a phenomenon currently attributed to something unknown:

- A galaxy rotating inside a medium that swirls along with it experiences stress from that swirl. Stress gravitates. The extra pull that standard cosmology assigns to a halo of undetected particles, in its own terms, might instead be the vacuum's own response to being rotated.
- A medium with internal pressure that can change character over time can drive expansion without a constant pasted into the equations by hand. What is now called dark energy would be the Ocean in a particular state rather than a number in a table.
- And — this is the part developed furthest here — a medium has a smallest meaningful size. Below some scale, it cannot be disturbed in a localized way at all. That gives a principled distinction between what counts as a localized pattern and what counts as a ripple.

The ambition is that dark matter and dark energy stop being two mysteries and become two behaviours of one thing.

What this document is not

It is not a theory. It is a proposal with its homework listed.

The equations below are consistent in form, but consistency in form is cheap. A physical theory has to survive conservation laws, stability analysis, the recovery of tested results in the appropriate limits, and quantitative agreement with observations that were not used to build it. None of that has been done here. Section 11 is an itemized list of the work required, and several items on it could terminate the framework outright. They are stated plainly rather than buried, because a proposal that cannot specify how it would fail is not worth the paper it occupies.

How this document came to exist

The framework began as private speculation, written to a standard that required only that the pieces appear to fit. That is a much weaker standard than the one it has since been held to, and the gap between the two turned out to be where the interesting work was.

Four revisions have followed, and they share a shape.

The first corrected a structural error. In the original formulation the vacuum's viscous response *replaced* the Einstein tensor — the geometric heart of General Relativity. This cannot work: a mathematical identity guarantees that the Einstein tensor is conserved, and substituting something else for it breaks the conservation of energy and momentum. The fix was to keep the geometry intact and add the vacuum's stress alongside it rather than in its place.

The second removed a number. The original drew the line between matter and vacuum at 0.511 MeV — the rest energy of the electron. Above the line, matter; below it, vacuum. This was an acknowledged placeholder, and it had three problems: the value was imported rather than derived, it was observer-dependent (two observers moving relative to one another would disagree about which side of the line a given thing fell), and it was an abrupt cutoff where physics prefers continuity.

The replacement drops the number entirely and asks instead how small a disturbance the medium can hold. Every medium has such a scale — in a laboratory superfluid it is called the healing length, the shortest distance over which the medium can bend without tearing. Disturbances larger than it spread out and remain part of the medium. Disturbances smaller than it stay put, as localized patterns. The matter/vacuum distinction becomes a property of the Ocean rather than a line drawn across it.

The result worth noting is what happened next. Calibrating that scale to present conditions puts it at roughly 386 fm, which is exactly the electron's own characteristic length, the scale below which it cannot be pinned down. It is a threshold, not a radius: probed directly, the electron shows no size at all down to a scale a hundred million times smaller. The electron, on this account, is simply the lightest pattern the Ocean can hold in place — and everything lighter, photons and neutrinos included, is necessarily a ripple instead. The number that had been an unexplained input came back out as a consequence. That is the kind of exchange that makes a framework worth continuing.

The third responded to news from outside. In September 2026 a proof was announced that a fluid with internal friction can, under smooth external stirring, tear itself into a singularity in finite time — a result about ordinary fluid dynamics with no connection to this framework whatsoever. It remains unverified and does not transfer directly to a relativistic medium. But it removed an assumption this framework had been resting on without noticing: that viscosity is enough to keep a continuum well-behaved forever.

The repair was already present. The smallest-scale limit introduced in the second revision is precisely what stops a disturbance concentrating without bound — but only if the medium resists compression rather than welcoming it. That condition had never been written down. It is now a stated requirement, and if the Ocean turns out to have the other sign, the framework fails as written.

The fourth is the one this version records, and it is a change of mind. The first revision had kept Einstein's geometry intact and set the Ocean beside it, and every version since had carried that arrangement forward: a stage, and a medium placed upon it. The stage was never argued for. It was inherited, from the habit of treating spacetime as the thing that exists and fields as things that happen in it. Asked directly what the stage was made of, the framework had no answer, and a medium that fills everything leaves no room for a second structure to occupy.

So the stage is removed. There is only the Ocean. Einstein's equation stays, because it describes what is measured with extraordinary precision, but its standing changes: it is now the description of how the medium's state is experienced by patterns made of that medium, not a separate structure the medium sits in. Gravity, on this account, is the medium's response to concentrated energy. That claim brings obligations the earlier arrangement did not carry, and they are set out in the technical sections; the heaviest is to derive the geometry rather than assume it.

Each revision has the same form: something chosen by hand is removed, and something forced by the structure takes its place. A substituted tensor, then an imported number, then an unexamined assumption,

then an inherited stage. That a framework can absorb that process repeatedly is not evidence that it is true. It is evidence that it is the right kind of thing to keep testing — which is a lower bar, and the only one this document claims to clear.

A note on method, since it is unusual enough to be worth recording. The revisions above were worked out in dialogue with a large language model, used as an interlocutor rather than an author: a way of being argued with at the speed of thinking. There is a small irony in the third revision, where the result that forced the change was itself produced by a machine, formally verified by another, and is still waiting on human confirmation. The fourth went the other way: the position had been the author's from the start, and the dialogue's work was to notice that the document had never actually said it.

What would kill it

Four things, named here so that a reader does not have to hunt for them.

The scale gap. For the Ocean to explain galactic rotation it must be able to act across thousands of light years. For it to place the matter boundary at the electron it must have structure at the scale of a hundredth of an atom. These two requirements differ by about thirty-two orders of magnitude. They are set by different parts of the mathematics, so this is not an outright contradiction — but a framework that merely tolerates such a gap has not explained anything, and this is the most serious unresolved problem in the document.

A route out now exists, and it is recorded as a route rather than a result. A hurricane is hundreds of miles across, but that size is not a property of air. It belongs to the storm: to how much energy and spin were gathered in one place. Air itself has only small-scale properties. If the same holds here, the galactic scale is not built into the Ocean at all. A galaxy is not an object that stirs the medium; it is a region of the medium carrying excess energy and excess current, of which the visible stars are the dense middle and the wide slow flow is the outskirts. The Ocean would then need only one built-in size, the small one, together with the ability to carry very long, slow disturbances at almost no cost. Laboratory superfluids have that ability for a known reason, a symmetry of the underlying field. The price is that the field must be a richer object than the one written down below, and that the calculation showing such a disturbance reproduces observed galaxies has not been done. Until it is, the gap stays on this list.

The route does not make the gap smaller. A galaxy really is about 10^{32} times larger than the Ocean's smallest structure, just as a hurricane is about 10^{14} times wider than the spacing of air molecules. What the route changes is whether that ratio needs explaining. If the large scale belongs to the disturbance and not to the medium, it is the size of a storm, not a second setting of the air.

The sign. As above: if the Ocean's self-interaction has the wrong sign in its present state, nothing arrests collapse and the framework is not viable in its present form. This is a single calculation away from being settled either way. Settling it for the Ocean's present state does not settle it for every time and place. The medium resists compression only in some of its possible states; in others it may not, and the earliest moments of the present beat or distant regions could have been in such a state. The rules do not change from place to place, but where the medium sits within them can.

The preferred frame. Describing the vacuum as a medium gives it a state of rest, and a state of rest is something Einstein spent a career removing from physics. Real superfluids do exactly this, so it is not absurd — but the experimental limits on any such effect are extraordinarily tight, and the framework must show it stays inside them.

The measured vacuum. Empty space has already been measured doing things. It screens electric charge, it scatters light off light, and in very strong magnetic fields it splits light by polarisation, as a crystal does. Standard physics predicts these effects by other means, in some cases to many decimal places. If the Ocean is real, these are its properties, and it must reproduce the same numbers. Agreement would be strong support. Any disagreement would be fatal.

To which should be added a possibility that no calculation will resolve: the agreement between the Ocean's smallest scale and the size of the electron may simply be a coincidence. The Standard Model

already accounts for the electron's mass by other means. Whether these are two descriptions of one fact or two unrelated facts that happen to agree is, at present, unknown.

What follows is the technical statement of all of this. It is written for a reader comfortable with tensor notation, but the argument it makes is the one above: that the emptiness between things may not be empty, and that a great deal of what we currently call unknown may be the behaviour of a single medium that has been sitting in plain view the whole time.

Conventions. No independent particulate entities are posited. An excitation of the medium that is localized in the sense of Section 7 is called a *localized excitation*; one that is not is a *coherent excitation* or a *flow*. "Matter" and "the matter sector" denote the localized population of the one medium, never a second substance. Particle-like behaviour — the exchange of energy, momentum and charge in whole units at a single point — is treated as a property of how excitations of a quantum field interact, not as evidence of objects (Section 7). The word "particle" appears only when describing other work, and is marked as that work's term. The framework is four-dimensional throughout: indices μ, ν run over one dimension of time and three of space, and the Ocean is a medium in all four. Time is part of the medium's description, not a clock standing outside it.

Glossary. The following terms are used with fixed meanings throughout.

The Ocean
The single continuous medium. It has no edges and no outside, and it includes every beat. Nothing in the framework exists apart from it.

Pattern
Any excitation of the Ocean, localized or not. Matter and light are patterns; so is a galaxy.

Localized excitation
A pattern localized in the sense of Section 7; what other work calls a particle.

Coherent excitation, flow
A pattern that is not localized.

Matter, matter sector
The localized population of the Ocean, never a second substance.

The Heartbeat
The whole sequence of beats.

A beat
One expansion and contraction of the Ocean, from one beat transition to the next.

The present beat
The beat we are in, and the only one any observation can reach.

Beat transition
The change of state between one beat and the next. Never a phase change within a beat.

The big bang
The most recent beat transition, as seen from within the present beat. Not the beginning of the Ocean.

Epoch
A phase within a beat, such as the radiation, matter or dark-energy epoch. Never a beat.

The observable universe
The part of the present beat that observation can reach. Neither the whole present beat nor the Ocean.

"Universe" is not used on its own, since what it names is redefined here; it appears only in "the observable universe" and in reporting other work. "Cycle", like "particle", appears only when reporting other work. "Cyclic" in the subtitle names the established category of cosmology to which the Heartbeat belongs.

1 The Field Equation

The governing equation for the "Ocean" metric is formulated as Einstein geometry coupled to a dynamic scalar vacuum medium. The central revision relative to the original formulation is that the viscous

response is treated as a *stress contribution* rather than as a replacement for the geometric Einstein tensor:

$$G_{\mu\nu} + P(\phi)\, g_{\mu\nu} \;=\; \kappa(\phi)\Big[\, T^{\text{matter}}_{\mu\nu} + T^{\phi}_{\mu\nu} + \Pi_{\mu\nu}\,\Big], \qquad \kappa(\phi) = \frac{8\pi G(\phi)}{c(\phi)^4}. \tag{1}$$

Here $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu}$ remains the geometric Einstein tensor; $P(\phi)$ is the dynamic vacuum pressure; $T^{\phi}_{\mu\nu}$ is the stress-energy of the scalar field; and $\Pi_{\mu\nu}$ is its hydrodynamic dissipative stress.

Status of the equation. Equation (1) is retained as the working equation of the framework, but its standing has changed. Earlier versions treated $G_{\mu\nu}$ as a fundamental structure with the medium added to it. Here the medium is primary: the geometry on the left-hand side is the effective description of how the medium's state is experienced by excitations of the medium, and the right-hand side records the medium's own stresses. The equation is kept because Einstein's description is correct to the precision it has been tested; the framework's claim is to explain why it holds (Section 2.5), and deriving it from the dynamics of the medium is the central task that claim creates (Section 11).

2 Tensor Definitions

2.1 Geometric Einstein tensor

The geometric sector retains the standard covariant structure of General Relativity:

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}. \tag{2}$$

This separation is essential because the contracted Bianchi identity gives $\nabla^{\mu} G_{\mu\nu} = 0$. Any additional vacuum stresses must therefore satisfy the corresponding total conservation requirement (Section 9).

2.2 Scalar vacuum stress-energy $T^{\phi}_{\mu\nu}$

The vacuum is represented by a dynamical scalar field rather than a purely phenomenological density. Let

$$X \equiv -\tfrac{1}{2}\, \nabla_{\alpha}\phi\, \nabla^{\alpha}\phi, \tag{3}$$

and let $L(\phi, X)$ be the scalar-field Lagrangian. Its stress-energy tensor is

$$T^{\phi}_{\mu\nu} \;=\; L_{,X}\, \nabla_{\mu}\phi\, \nabla_{\nu}\phi \;+\; L\, g_{\mu\nu}. \tag{4}$$

This term gives the Ocean a genuine dynamical energy density and pressure. The specific choice of L determines whether the field behaves as a conventional scalar, a k-essence-like medium, or a more specialized superfluid effective theory. Section 7 depends on this choice directly.

A candidate refinement. The candidate route for the scale hierarchy (Section 7.7, item (5)) requires more structure than a single real scalar. Take the medium to be a complex field with a global $U(1)$ symmetry,

$$\Phi = \rho\, e^{i\theta}, \qquad \theta \;\to\; \theta + \text{const}, \tag{5}$$

and identify the ϕ of the preceding expressions with the phase θ. The shift symmetry forbids a potential for θ, so that $L(\phi, X) \to L(X)$ and the long-wavelength excitation is a gapless Goldstone phonon, as in a laboratory superfluid. The modulus ρ is the natural home of the self-interaction of Section 9.1, and its excitation is heavy.

This is a condition the route depends on, not an adopted change, and it has a cost that propagates. An exact shift symmetry forbids any dependence on the value of θ itself. The state-dependent quantities $P(\phi)$, $\kappa(\phi)$, $\xi(\phi)$, $\eta(\phi)$ and $\zeta(\phi)$ would have to be re-expressed as functions of the modulus ρ or of X, and the beat transitions of the Heartbeat would become transitions of ρ. The same applies to the

quartic potential of Section 9.1, which becomes a potential for ρ, and to the mass-gap entry of the table in Section 7, which vanishes identically for θ. The route also brings the framework's field content closer to the complex-field description of Huang and collaborators [1] and to the phonon description of Berezhiani and Khoury [2]; the distinctive claim of Section 10, the partition with a retained fundamental geometry, is unaffected.

2.3 Vacuum viscosity tensor $\Pi_{\mu\nu}$

The viscous response represents the dissipative stress response of the scalar medium to local flow. For a relativistic effective fluid, a first-order constitutive form is

$$\Pi_{\mu\nu} = -2\,\eta(\phi)\,\sigma_{\mu\nu} \;-\; \zeta(\phi)\,\theta\,h_{\mu\nu}, \tag{6}$$

where $\eta(\phi)$ is the shear viscosity, $\zeta(\phi)$ the bulk viscosity, $\theta = \nabla_\alpha u^\alpha$ the expansion scalar, $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ the spatial projector orthogonal to the medium four-velocity u^μ, and $\sigma_{\mu\nu}$ the shear tensor.

In regions with coherent vorticity or differential flow, $\Pi_{\mu\nu}$ can generate additional gravitational stress. The model therefore investigates whether the vacuum response can reproduce effects conventionally assigned to dark-matter halos. The term *rotational support* is used here as a hypothesis about the gravitational effect of the vacuum stress response; viscosity itself is not assumed to behave as additional localized mass.

2.4 Scalar pressure $P(\phi)$

$P(\phi)$ replaces a fixed cosmological constant as the phenomenological vacuum-pressure contribution. It is a function of the state of the underlying scalar field rather than a universal constant:

$$\Lambda g_{\mu\nu} \;\longrightarrow\; P(\phi)\,g_{\mu\nu}. \tag{7}$$

The *Heartbeat* hypothesis proposes that the scalar field can evolve through alternating dynamical phases. An effective positive-pressure / negative-pressure transition may then produce expansion ("diastole") and contraction ("systole") phases. A complete model must derive this behavior from a field potential or equation of state rather than prescribing an oscillating pressure by hand.

2.5 Effective coupling $\kappa(\phi)$

The coupling between the vacuum state and ordinary stress-energy is allowed to depend on the local scalar density:

$$\kappa(\phi) = \frac{8\pi G(\phi)}{c(\phi)^4}. \tag{8}$$

$G(\phi)$ represents a field-dependent effective gravitational coupling. The quantity $c(\phi)$ is treated initially as an effective propagation speed associated with the vacuum medium. If it is interpreted as a genuine variable causal speed, the theory must specify the resulting causal structure and recover the observed local value c_0 in the appropriate limit.

The refractive reading. The account of gravity adopted here is refractive. A wave in a medium whose propagation speed varies curves toward the region where it travels more slowly. If concentrated energy lowers the local propagation speed of the Ocean, light curves inward, and so does every localized excitation, each of which is a pattern the medium keeps re-making. Where the medium is slower on one side, the re-making is uneven and the pattern is re-made slightly further toward the slow side. Nothing pulls; trajectories are turned. This is consistent with the absence of any felt force in free fall. In the weak field it reproduces the standard equivalence between gravitational light bending and a refractive index

$$n(r) \;\simeq\; 1 + \frac{2GM}{rc^2}. \tag{9}$$

Refraction alone does not reproduce the full weak-field and post-Newtonian behaviour — perihelion advance, frame dragging, gravitational radiation — and extending it to do so is the test this reading must pass (Section 11). Three requirements are decisive. Every excitation must respond to the medium's state in exactly the same proportion, since all bodies and light are observed to fall alike to extraordinary precision; here that equality must follow from everything being made of the same medium. The response must slow time and stretch space in equal measure: the measured deflection of starlight by the Sun is twice what slowing alone would give, and the second half comes from the spatial part. And disturbances of the response itself, gravitational waves, must travel at the speed of light, as observed when a neutron-star merger's gravitational and light signals arrived together. The alternative reading of $c(\phi)$, a genuine variable causal speed, is retained as the fallback.

3 Scalar-Field Dynamics

The field ϕ must have an independent equation of motion. A generic covariant form is

$$\Box\phi \;=\; \frac{\partial V_{\text{eff}}}{\partial \phi} \;+\; C(\phi, T, \nabla\phi, \ldots)\,, \tag{10}$$

where $V_{\text{eff}}(\phi)$ is an effective potential and C represents permitted couplings to matter, curvature, or hydrodynamic variables. The potential and coupling structure determine the vacuum phases, stability, characteristic sound speed, and possible Heartbeat behaviour.

4 The Ocean Interpretation

The framework interprets the observable universe as a dynamical region of a deeper vacuum medium. The Ocean is not required to begin at the big bang. Instead, the big bang is interpreted as a beat transition: a change of state of the medium, a phase transition, or a rapidly expanding solution of the underlying field equations.

The observable universe is therefore not treated as the ontological boundary of reality. It is the portion of the deeper continuous vacuum field accessible to observation. This distinction is a conceptual premise of the framework and must ultimately be connected to a mathematically well-defined global solution.

5 Cosmological Heartbeat

The proposed *Heartbeat* cosmology associates cosmic expansion and contraction with the evolution of ϕ. A schematic beat is

ϕ-state transition $\rightarrow$ pressure reversal $\rightarrow$ expansion $\rightarrow$ field evolution $\rightarrow$ contraction $\rightarrow$ new phase.

The model does not require that each beat create the underlying vacuum field. Instead, each *beat* (Section) may be a dynamical state of the same deeper medium.

Two kinds of statement are kept apart in what follows. Section 5.1 concerns the present beat, which is the only one any observation can reach. Section 5.2 concerns the sequence of beats and the transitions between them, about which no observation made within a beat can speak directly.

5.1 Within the present beat

Everything known about time is known from within the present beat. Every observer and every instrument is a pattern of the Ocean in this beat, and every observation is of this kind. Here time is the pace at which the Ocean re-makes its patterns, measured by clocks that are themselves such patterns. Durations are meaningful and measurable, but local: clocks run at different rates in different places, as General Relativity describes and as the account of Section 2.5 requires.

The pace of the beat is not fixed. While matter dominated, its gravity slowed the expansion; distant supernovae show that the expansion is now accelerating [3, 4]. There are further indications that the acceleration is itself evolving: baryon acoustic oscillation measurements from the DESI survey, combined with the microwave background and supernova samples, prefer a dark-energy component that changes over time to a constant one, at roughly 3 to 4σ depending on the supernova sample, short of the level conventionally required for discovery [5]. In this framework such variation is the Ocean's state changing through $P(\phi)$, not a departure from a constant. One caution follows from the same data. The preferred evolution crosses the boundary $w = -1$, which a single scalar field with a standard kinetic term cannot readily do; the survey's authors note that multiple fields, additional internal degrees of freedom or a non-minimal coupling can evade the difficulty [5]. If the evolution is confirmed, the Ocean's pressure sector would need such structure, and the complex field of Section 2.2 and the coupling $\kappa(\phi)$ are the candidates already present.

5.2 Across beats

Nothing in this subsection is observable from within the present beat. It states what the framework permits, not what has been found, and the two subsections should not be read into one another.

No clock survives the transition between beats. The patterns that keep time are diluted or remade, and the Ocean's state, including ξ, may differ from one beat to the next. Durations therefore cannot be compared from beat to beat, and "faster" or "slower" across beats can only mean that different beats pass through different sequences of states. The framework requires no repetition: each beat may differ in its expansion, its reset and its ξ. Across beats the Heartbeat is counted, not timed.

Whether such a sequence can avoid singularities, entropy accumulation and instabilities is an open mathematical question. Because the stiffness that arrests collapse depends on the medium's state (Section 9.1), the answer may differ between phases of a beat: a contraction that drives the medium out of its stiff state need not be arrested by the mechanism that protects the present epoch.

The nearest established precedent is the cyclic model of Steinhardt and Turok [6], in which a scalar field rolling in a potential drives an endless sequence of expansions and contractions, each big bang being a transition from a preceding contraction with finite temperature and density. Two of its answers bear directly on the questions above. Entropy does not accumulate from cycle to cycle because a long phase of accelerated expansion, the present dark-energy epoch, dilutes the matter, radiation and black holes of the previous cycle before contraction begins. And the nearly scale-invariant density fluctuations are generated during the slow contraction rather than by inflation, with the distinguishing prediction of no long-wavelength primordial gravitational waves. Both are available to the Heartbeat in principle. The difference lies at the bounce: in that model the transition is motivated by colliding branes along an extra spatial dimension, with the four-dimensional field theory carrying a matching rule supplied by hand. The premise of this framework admits no space outside the Ocean, so the bounce must instead come from the medium itself, through the healing-length floor and the state dependence of its stiffness (Section 9.1).

One consequence of this route can be tested from within the present beat. If the Heartbeat seeds the density fluctuations of the present beat during the preceding slow contraction, as in the Steinhardt–Turok model, rather than by inflation, it predicts essentially no long-wavelength primordial gravitational waves. The present upper limit on their amplitude relative to the density fluctuations, from Planck combined with BICEP2/Keck Array data, is $r < 0.056$ [7], and forthcoming microwave-background experiments will test it much further. A detection of primordial gravitational waves would count against a contraction-seeded Heartbeat; continued non-detection would leave it viable. This route also avoids a difficulty noted for inflation in this framework: the field values that inflation requires lie far above the scale set by ξ_0.

6 Galactic Dynamics

In the galactic regime, the hypothesis is that spatial gradients, vorticity, and coherent flow of the vacuum field generate an additional effective stress. The observable circular velocity would then be determined

by the combined baryonic and vacuum-field gravitational response:

$$v_c^2(r) \;=\; v_b^2(r) \;+\; v_{\phi,\mathrm{eff}}^2(r). \tag{11}$$

Here v_b is the baryonic contribution and $v_{\phi,\mathrm{eff}}$ denotes the effective contribution produced by the scalar field and its stress response. The goal is not to insert an arbitrary halo profile, but to derive $v_{\phi,\mathrm{eff}}(r)$ from ϕ, $\Pi_{\mu\nu}$, and the field equations.

Rotation is a special case. The relation above applies to rotationally supported disks. Galaxies also carry random motion, inflow and outflow, vertical oscillation, bars and spiral patterns, and motion through their surroundings. Pressure-supported systems with little net rotation — elliptical galaxies and dwarf spheroidals — follow the same radial acceleration relation as disks [8]. Whatever links the vacuum contribution to the baryons therefore cannot depend on rotation, and the framework must apply to pressure-supported galaxies as well.

One structure, not two. The decomposition above is bookkeeping and should not be read as a statement that a galaxy acts on the medium from outside. In this framework a galaxy is a region of the Ocean carrying excess localized energy and elevated current. Its baryonic content is the $\mathscr{Q} \gg 1$ population of that region and the extended flow is the $\mathscr{Q} \ll 1$ population, in the sense of Section 7; the two are the dense interior and the diffuse exterior of a single solution of the field equations. This differs from treatments in which galaxies are external sources perturbing the superfluid [1].

Two consequences follow. First, the kiloparsec scale is a property of the individual solution, fixed by the full state of the disturbance — its energy, its organised and random flow, and its exchange with the surrounding medium — and need not be a parameter of the medium; this is the basis of the candidate route of Section 7.7, item (5). Second, if the solution is fixed by its baryonic content — the question posed by the Size task of Section 11 — the baryonic distribution determines $v_{\phi,\mathrm{eff}}(r)$ with no freedom adjusted galaxy by galaxy. The observed radial acceleration relation has that form: across rotationally supported galaxies the total acceleration is fixed by the baryonic acceleration alone, with little or no intrinsic scatter and no sign of a second parameter [9, 8]. That is a target to be derived, not a result.

It also sharpens the exposure recorded in Section 10. A contribution fixed by the baryons is precisely the kind that risks being parallel to the baryonic gravitational field, and a parallel contribution is excluded by Milky Way vertical kinematics [10, 11]. The single-structure reading therefore needs a vacuum contribution whose magnitude is set by the baryons while its direction is not slaved to theirs. Whether the field equations permit that is not known.

The vacuum contribution is a flow. The decomposition $v_c^2 = v_b^2 + v_{\phi,\mathrm{eff}}^2$ reduces the vacuum's response to a single radial quantity. In the framework it is a flow in time and space together, not a single radial number. Momentum conservation means that any local acceleration of the medium produces shear against neighbouring regions and, for a disturbance moving through its surroundings, a wake; the shear term $\sigma_{\mu\nu}$ of $\Pi_{\mu\nu}$ is where this enters. Such a flow is not constrained to lie parallel to the baryonic gravitational field. A wake is aligned with the direction of motion, not with the baryonic field. This is one way the vacuum contribution could satisfy the parallelism criterion above, and it is untested. It also implies an asymmetry: the vacuum contribution of a moving galaxy should be offset or lopsided relative to its direction of motion. Colliding clusters, in which the inferred extra mass follows the galaxies rather than the slowed gas, are the hard case for any such picture.

7 The Localization Threshold

7.1 Why a fixed threshold fails

The earlier formulation partitioned energy between sectors by a declared inequality:

$$E > 0.511\,\mathrm{MeV} \;\Rightarrow\; \text{localized matter sector}, \qquad E \le 0.511\,\mathrm{MeV} \;\Rightarrow\; \text{scalar-fluid density } \phi.$$

This has three defects, only the first of which is usually noticed.

(1) **It is an unexplained parameter.** The value is imported from the electron rest mass, which the framework does not derive.

(2) **It is not covariant.** E is frame-dependent. A boost changes which sector a given excitation belongs to, so the partition is observer-dependent — unacceptable in a theory built on the covariant field equation (1).

(3) **It is discontinuous.** A step function requires an exchange term coupling $T^{\text{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$. Combined with the variable coupling $\kappa(\phi)$, this makes the conservation conditions of Section 9 essentially impossible to satisfy.

The revision removes all three by replacing the declared threshold with a property of the vacuum field's own excitation spectrum.

7.2 The excitation spectrum of ϕ

Perturb the field about a background configuration,

$$\phi(x) = \bar{\phi}(x) + \delta\phi(x), \tag{12}$$

and let u^μ be the four-velocity of the vacuum medium with spatial projector $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$. Expanding the Lagrangian $L(\phi, X)$ of Section 2.2 to second order in $\delta\phi$ gives a quadratic action of the form

$$S_2 = \frac{1}{2}\int \sqrt{-g}\left[\mathscr{K}\left(u^\mu\partial_\mu\delta\phi\right)^2 - L_{,X}\,h^{\mu\nu}\partial_\mu\delta\phi\,\partial_\nu\delta\phi - V''_{\text{eff}}\,\delta\phi^2 - \mathscr{G}\left(h^{\mu\nu}\partial_\mu\partial_\nu\delta\phi\right)^2\right], \tag{13}$$

with $\mathscr{K} \equiv L_{,X} + 2XL_{,XX}$. The final term is a gradient-stiffness (quantum-pressure) operator. In a superfluid effective theory it arises from the same expansion; here it must be *derived* from the chosen L or from an underlying superfluid EFT, not assumed (see Section 7.7).

Varying (13) and evaluating in the medium rest frame yields the dispersion relation

$$\boxed{\omega^2(k,\phi) \;=\; \Omega^2_{\text{gap}}(\phi) \;+\; c_s^2(\phi)\,k^2 \;+\; \frac{\hbar^2}{4m_\star^2(\phi)}\,k^4} \tag{14}$$

7.3 The three characteristic scales

Every coefficient in (14) is fixed by the Lagrangian rather than chosen:

Quantity	Definition	Physical meaning
Sound speed	$c_s^2 = L_{,X}/\mathscr{K}$	Propagation speed of coherent Ocean disturbances
Mass gap	$\Omega^2_{\text{gap}} = V''_{\text{eff}}(\bar{\phi})/\mathscr{K}$	Curvature of the effective potential at the local vacuum state
Effective inertia	$m_\star = m_\star(\mathscr{G}, \mathscr{K})$	Stiffness of the field against short-wavelength bending

From these follows the single scale that carries the whole section — the *healing length*:

$$\xi(\phi) \;=\; \frac{\hbar}{m_\star(\phi)\,c_s(\phi)}. \tag{15}$$

ξ is the shortest distance over which the vacuum field can vary without the gradient term dominating. It is the Ocean's localization scale: a threshold, not a granularity (Section 7.7, item (7)).

7.4 The localization criterion

Define the dimensionless quantity

$$\mathscr{Q}(\phi,k) \;\equiv\; \xi^2(\phi)\, h^{\mu\nu} k_\mu k_\nu. \tag{16}$$

Because $h^{\mu\nu}$ projects onto the rest frame of the medium, $\mathscr{Q}$ is constructed covariantly from the excitation's four-momentum and the medium's own state. It replaces the frame-dependent E of the old formulation.

$$\mathscr{Q} \ll 1 \implies \omega \approx c_s k \qquad \text{phononic branch; delocalized; contributes to } T^{\phi}_{\mu\nu}, \tag{17}$$

$$\mathscr{Q} \gg 1 \implies \omega \approx \hbar k^2/2m_\star \qquad \text{localized branch; contributes to } T^{\text{matter}}_{\mu\nu}. \tag{18}$$

The crossover sits at $k_c = 1/\xi(\phi)$, and the corresponding energy is

$$E_c(\phi) = \hbar\,\omega(k_c) \;\approx\; \alpha\, m_\star(\phi)\, c_s^2(\phi), \qquad \alpha = \mathscr{O}(1). \tag{19}$$

The partition is smooth, not switched. Define a monotonic weighting $f(\mathscr{Q})$ with $f(0) = 0$ and $f(\infty) = 1$ — for example $f = \mathscr{Q}/(1+\mathscr{Q})$ — and split the mode integral:

$$T^{\text{matter}}_{\mu\nu} = \int d^3k\, f(\mathscr{Q})\, t_{\mu\nu}(k), \qquad T^{\phi}_{\mu\nu} = \int d^3k\, \big[1 - f(\mathscr{Q})\big]\, t_{\mu\nu}(k). \tag{20}$$

Since both sectors are populations of the *same* field, their sum is conserved identically. No exchange term is required, and the consistency conditions of Section 9 are no longer obstructed by the threshold.

7.5 Recovering 0.511 MeV as a calibration

Requiring $E_c(\phi_0) \approx 0.511\,\text{MeV}$ in the present vacuum state fixes

$$\xi(\phi_0) = \frac{\hbar c}{E_c} \approx 386\,\text{fm}, \tag{21}$$

which is the reduced Compton wavelength of the electron. This is a derived consequence rather than a coincidence. A localized excitation of rest energy mc^2 has Compton wavelength $\bar{\lambda}_C = \hbar/mc$; it can localize within the medium only if $\bar{\lambda}_C \lesssim \xi_0$, which requires

$$m \;\gtrsim\; \frac{\hbar}{c\,\xi_0} \;=\; m_e. \tag{22}$$

The electron is therefore the lightest excitation the present vacuum can hold as a localized pattern. Anything lighter has a Compton wavelength longer than the Ocean's localization scale and cannot be localized in it — it is necessarily a coherent field excitation. This is offered as the framework's explanation for why photons and neutrinos behave as field phenomena while the electron is the floor of the localized sector. The value $0.511\,\text{MeV}$ is demoted from axiom to measurement: it is the readout of $\xi(\phi_0)$, not an input.

7.6 Consequences

The threshold is local and dynamical. $E_c = E_c(\phi(x))$, so regions in different vacuum states have different matter thresholds. This connects the present section to the Heartbeat cosmology: a phase transition of ϕ is also a transition in what the vacuum is *able to hold* as matter. Where ξ grows, the localization floor rises and previously localized states are no longer supported.

It predicts a varying electron mass. If ξ evolves, so does the effective m_e. This is a genuine observational handle — and a genuine constraint.

It introduces a preferred frame. The dispersion relation (14) is stated in the rest frame of u^μ. This is honest — real superfluids do exactly this — but it is a departure from exact local Lorentz invariance and must be reconciled with precision tests.

Particle-like behaviour does not require smallness. An excitation appears particle-like because it exchanges energy, momentum and charge in whole units and at a single point. Both properties follow from the medium being a quantum field with local couplings, independently of the localization scale. The healing length is a threshold for localization, not a radius, and above it the term used here is *excitation* or *coherent structure*, whose size is set by circumstance.

7.7 Open items specific to this section

(1) **Derive the gradient-stiffness term** $\mathscr{G}$ from a covariant action rather than importing it from the non-relativistic Bogoliubov form. Without this, $m_\star$ and hence ξ are not defined by the theory.

(2) **Stability.** Require $L_{,X} > 0$ and $\mathscr{K} = L_{,X} + 2XL_{,XX} > 0$; otherwise $c_s^2 < 0$ (gradient instability) or the kinetic term flips sign (ghost).

(3) **Lorentz-violation bounds.** Show $c_s \to c_0$ in the local limit to the precision demanded by clock-comparison and photon/neutrino time-of-flight tests, and demonstrate that the k^4 term is suppressed at accessible energies. The k^4 term becomes significant for wavelengths near ξ_0, that is at energies near 0.5 MeV. Photons from distant gamma-ray bursts show no energy-dependent arrival times at energies far above this, so the k^4 term cannot act on the electromagnetic sector, nor on electrons, at the scale ξ_0. Either it is confined to excitations of ϕ, with a stated reason, or the framework is excluded. Relatedly, a gamma-ray photon has a wavelength far below ξ_0, so the criterion (16) must be stated as applying to excitations of ϕ only, or it would classify such photons as localized.

(4) **Varying-m_e bounds.** Quasar absorption spectra constrain drift in m_p/m_e at roughly the 10^{-5}–10^{-6} level over cosmological time; the Oklo natural reactor constrains it locally. Any Heartbeat evolution of ξ must be negligibly slow over the epochs these measurements probe, and largely confined to beat transitions.

(5) **The scale hierarchy.** Galactic-scale effects require the scalar's Compton range to be of order a kiloparsec, implying $\Omega_{\rm gap} \sim 10^{-26}\,\mathrm{eV}$. But $\xi_0 \approx 386\,\mathrm{fm}$ implies a stiffness scale near 0.5 MeV. These differ by roughly thirty-two orders of magnitude. They are set by different operators — $V''_{\rm eff}$ versus $\mathscr{G}$ — so this is not immediately inconsistent, but a credible model must *explain* the separation rather than merely tolerate it. This is the most serious open problem in the revised section.

Candidate route. The estimate $\Omega_{\rm gap} \sim 10^{-26}\,\mathrm{eV}$ assumes that the kiloparsec scale is intrinsic to the medium. On the single-structure reading of Section 6 it is instead a property of the individual galactic solution. The requirement on the medium then weakens from a tuned value to an upper bound,

$$\Omega_{\rm gap} \lesssim 10^{-26}\,\mathrm{eV}, \tag{23}$$

which $\Omega_{\rm gap} = 0$ satisfies. A vanishing gap is protected, rather than tuned, if ϕ is the phase of a complex field with a $U(1)$ shift symmetry (5); a small nonzero gap from soft breaking of that symmetry would be technically natural. The medium would then carry one intrinsic length scale, ξ_0, and the hierarchy would be a ratio between a parameter of the medium and the size of a particular solution, which is not a naturalness problem. In the same construction the heavy modulus is the expected origin of the k^4 term in (14), which would bear on item (1).

None of this is established. The route trades the hierarchy for three obligations: to show that the symmetry survives the couplings of equation (1); to say what fixes the size of a galactic solution; and to pass the vertical-acceleration test in the form stated in Section 6. It also carries the cost recorded in Section 2.2. Until those are met the hierarchy stands as an open problem.

The route does not remove the ratio between ξ_0 and galactic scales; it changes it from a separation between two parameters of the medium into the size of a solution relative to the medium's one intrinsic length. A consistency check follows from the circulation quantum. A superfluid carries rotation in quantized vortices whose areal density is fixed by the local rotation rate and the mass of the condensate quanta (Feynman's rule). For quanta at the mass implied by ξ_0, and the rotation of the solar neighbourhood ($220\,\mathrm{km\,s^{-1}}$ at 8 kpc), the vortex spacing is of order 600 km, so the medium is smooth on galactic scales and the route is not contradicted by granularity. The estimate assumes $c_s \approx c$ and local rigid rotation. Because the medium then carries a length and no intrinsic acceleration, the acceleration scale $g_\dagger$ of the radial acceleration relation [9] must arise elsewhere, most plausibly from the cosmological background state.

(6) **Relation to the Standard Model.** The Higgs mechanism already accounts for m_e via a Yukawa coupling. The framework must state whether ξ_0 and the electron Yukawa are independent quantities that happen to agree, or whether one is meant to determine the other.

(7) **Size and spin of the localized excitation.** ξ_0 is the scale below which an excitation localizes, not its radius. Quantum electrodynamics has the same separation: the electron's reduced Compton wavelength limits how far it can be localized without pair creation, while scattering and its magnetic moment show no internal structure down to about 10^{-18} m. The framework must show that its localized excitation responds as a point to probes at that resolution. A second route exists: $\xi(\phi)$ is field-dependent, so the local healing length inside a concentrated excitation may be far smaller than ξ_0. That route is constrained from the other side, since the healing length must not shrink without bound in states where the medium is stiff, or the floor that arrests collapse in Section 9.1 disappears.

Separately, the electron has spin $\frac{1}{2}$. Excitations of a scalar field, real or complex, carry integer spin. The localized sector therefore requires structure that the present scalar formulation does not contain, and this applies to every fermion, including the neutrinos described above as field phenomena.

8 Limiting Regimes

Regime	**Required behavior**	**Physical objective**
GR / Solar System	$\phi \to \phi_0$; $G(\phi) \to G_0$; $c(\phi) \to c_0$; vacuum stresses negligible	Recover tested General Relativity
Galactic	$\Pi_{\mu\nu}$ and scalar gradients become significant	Explain rotation curves without dark matter in the particle form its proponents posit
Cosmological	$P(\phi)$ and $T^{\phi}_{\mu\nu}$ contribute on large scales	Generate expansion history and effective dark-energy behavior
Beat transition / Heartbeat	ϕ evolves between dynamically distinct states; $\xi(\phi)$ shifts	Provide a mechanism for the expansion and contraction of beats and for changes in the localization floor
High-energy	Mode occupation crosses $\mathscr{Q} \sim 1$	Test the derived localization threshold of Section 7
Strong field / black hole	Outward waves turned back at a horizon; collapse arrested near the healing length if $\lambda > 0$	Match horizon-scale imaging and merger ringdowns; determine whether a singularity forms

8.1 Black holes

In the refractive reading of Section 2.5, a black hole is a region where waves are slowed enough that an outward-directed wave is turned back. Laboratory analogues make this concrete. A fluid flowing faster than its own sound speed forms a horizon that sound cannot cross, and such a horizon has been made in an atomic Bose–Einstein condensate, where the spontaneous analogue of Hawking radiation was observed together with the entanglement of its pairs [12]. In that system the shortest length scale is the healing length $\xi = \hbar/mc_s$, the same construction as Section 7. General Relativity itself admits an exact description of a stationary black hole as space flowing inward through a flat background, reaching the speed of light at the horizon [13]. For a rotating black hole that description has the river twist rather than swirl. A draining vortex in water is nonetheless an analogue of a rotating black hole, and rotational superradiance — waves extracting energy from the rotation — has been observed in one [14].

Three questions follow.

The core. With $\lambda > 0$ the healing length resists compression, so collapse may terminate in a core of finite density rather than a singularity. That holds only while the compressed medium stays in a stiff state; if collapse drives it out of that state, the arrest is not guaranteed (Section 9.1). This is the question Section 5 asks of a contraction phase, posed for a single object.

Two horizons. The authors of the river model stress that its flat background has no observable meaning [13]. This framework retains $G_{\mu\nu}$ as fundamental and adds a physical medium, so a geometric horizon (where escape requires c) and a horizon of the medium (where escape requires c_s) need not coincide. If they differ, the shell between them requires account. If they always coincide, that suggests the geometry is the medium's own flow, which bears directly on the structural question of Section 10. A medium that literally flowed inward would also have to accumulate somewhere; the refractive reading avoids this, since the medium's state changes near the mass without net inflow.

Observation. Horizon-scale imaging and gravitational-wave ringdowns agree with General Relativity, so any departure must lie inside the horizon or below current sensitivity.

9 Conservation and Consistency Conditions

Because $\nabla^\mu G_{\mu\nu} = 0$, the complete right-hand side of (1) together with the pressure sector must satisfy a compatible conservation law. With variable $\kappa(\phi)$, the simple condition $\nabla^\mu T_{\mu\nu} = 0$ is generally not sufficient by itself. The scalar equation (10) and the matter coupling must be chosen so that the full theory is diffeomorphism-consistent.

The revised Section 7 assists here rather than obstructing: because $T^{\text{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$ are now weighted populations of a single field rather than two sectors joined by a step function, no discontinuous exchange term is introduced at the threshold.

9.1 Global regularity and the defocusing condition

A separate question, treated in earlier versions as a requirement that the medium remain smooth for all time, is whether it does. The framework cannot appeal to dissipation alone to secure this. Viscosity is conventionally expected to drain an energy cascade before it concentrates without bound, but that expectation is not a theorem: finite-time blowup constructions for the forced three-dimensional Navier–Stokes equations were announced in September 2026 and, while they remain subject to independent verification and do not transfer directly to a relativistic scalar medium, they remove the intuition that a viscous continuum is automatically globally regular. The shear and bulk viscosities $\eta(\phi)$ and $\zeta(\phi)$ of Section 2.3 therefore cannot be assumed to guarantee what they were implicitly relied upon to guarantee.

The mechanism available to this framework is not dissipation but the ultraviolet cutoff supplied by Section 7. The gradient-stiffness term in the dispersion relation (14) grows as k^4 and therefore dominates the cascade at $k \sim 1/\xi(\phi)$, arresting concentration at the healing length. This is the mechanism by which a laboratory superfluid remains regular where the corresponding classical fluid description does not.

That arrest is conditional on the sign of the self-interaction. Writing the relevant quartic term of the effective potential as

$$V_{\rm eff}(\phi) \supset \frac{\lambda}{4}\left(\phi^2 - v^2\right)^2, \tag{24}$$

the medium must be *defocusing*, $\lambda > 0$: compression must raise the energy and be resisted. The defocusing case is energy-coercive and the cutoff at ξ holds. In the focusing case $\lambda < 0$ the self-interaction rewards further compression, the cutoff does not arrest the cascade, and concentration accelerates instead. The sign is therefore a stability requirement of the same standing as $L_{,X} > 0$ and $\mathscr{K} > 0$, not a modelling preference. On the candidate route of Section 2.2 the potential belongs to the modulus ρ rather than to ϕ, and the argument carries over unchanged.

Regularity is a property of the state, not a guarantee. The sign of λ does not settle the matter for all times and places, because the stiffness that resists compression is the response of the medium *around the state it occupies*. With the Lagrangian held fixed, that response differs between states. Even the quartic form above, with $\lambda > 0$, has curvature

$$V''_{\rm eff}(\phi) = \lambda\left(3\phi^2 - v^2\right), \tag{25}$$

which is positive about the present vacuum $\phi = v$ and negative for $|\phi| < v/\sqrt{3}$. A medium driven into that region — by a beat transition, a contraction, or the core of a collapsing object — is no longer protected by the mechanism above. For this form the instability drives the medium back toward the stable state rather than into collapse; whether the full potential admits genuinely focusing states is open. The law is fixed; what varies is where the medium sits within it. No free function is added.

Four constraints follow. (i) Regions in different states are separated by boundaries that carry energy; walls of this kind within the observable universe are tightly excluded, so differing regions must lie beyond the horizon, in the past, or be separated without such walls. (ii) The microwave background is uniform to about one part in 10^5, which bounds how far the state could have varied across the observable universe at that epoch. (iii) A region that loses its stiffness collapses quickly and leaves relics, compact objects or primordial black holes, whose abundance is bounded. (iv) Where smoothness fails, the framework must state what follows — a bounce, a phase transition, or a genuine singularity at which it ceases to predict — and must tie each failure to a specified state of the medium, so that the possibility cannot be invoked case by case.

This section's argument rests on the framework's own structure. It does not depend on the blowup constructions cited above, which await independent verification.

A successful formulation should explicitly establish:

(1) covariance;

(2) total energy-momentum conservation;

(3) a well-posed scalar-field equation;

(4) absence of pathological ghosts or gradient instabilities;

(5) a stable low-density / weak-field limit;

(6) compatibility with precision tests of gravity and light propagation;

(7) a defocusing self-interaction about the present vacuum state, so that the healing-length cutoff of Section 7 arrests rather than accelerates concentration there, together with an account of the states in which it does not (Section 9.1).

10 Relation to Existing Work

The proposal that the vacuum is a superfluid, and that dark matter and dark energy are two behaviours of it rather than two substances, is not new. This section states what is already established, so that the

framework's actual claim can be isolated from the parts of it that are inherited.

10.1 Two families of superfluid-vacuum programme

Existing work divides on whether spacetime geometry is fundamental.

Emergent gravity. Volovik's programme [15] treats the quantum liquid as prior to spacetime: the metric, Lorentz invariance and gauge structure appear as low-energy collective behaviour of the condensate. Zloshchastiev's logarithmic superfluid vacuum theory [16] derives an induced gravitational potential from the vacuum wavefunction and concludes that dark matter and dark energy are manifestations of the same object and therefore cannot be independent. That conclusion is the one reached here. The mechanism is not: Zloshchastiev distinguishes his approach explicitly from models that work by modifying the stress-energy tensor of the Einstein equations.

Volovik's programme also supplies an argument this framework can use. Inhabitants of a superfluid, built from its own low-energy excitations, would perceive an effectively relativistic world and could not detect the underlying medium directly [15]. The same argument would account for the precision of Lorentz invariance at accessible energies here. It explains what cannot be observed; it does not exempt the framework from effects that are observed (Section 10, deficiency (4)).

Modified source. Huang and collaborators [1] treat a complex vacuum scalar field as a cosmic superfluid within standard General Relativity, identifying its energy density with dark energy and departures from vacuum density with dark matter. That work also treats galaxies as external sources perturbing the superfluid, and obtains the galactic halo together with the creation of vortices by galactic rotation — the closest existing precedent for the galactic sector of Section 6. Berezhiani and Khoury [2] construct an effective field theory in which dark matter is a superfluid with a galaxy-scale coherence length, described by phonons rather than, in the standard model's terms, collisionless particles, recovering MOND phenomenology on galactic scales. That model has since been disfavoured by Milky Way data [11]: it over-predicts vertical accelerations of disk stars even while reproducing the rotation curve. Viscous cosmology [17] models the dark sector as a single fluid whose bulk viscosity produces a negative effective pressure and thereby mimics dark energy. The logotropic models of Chavanis [18] belong to the same family, yielding a density profile that falls as r^{-1} and offering an alternative to MOND that accounts for the Tully–Fisher relation.

On the cosmological side, the closest structural precedent is the cyclic model of Steinhardt and Turok [6] (Section 5). It, too, is a four-dimensional scalar field theory in which one field accounts for dark energy, the reversal from expansion to contraction, and the transition between cycles, and it couples that field to matter through a field-dependent factor. Its authors note that this coupling is a modification of General Relativity and produces a fifth force violating the equivalence principle unless the coupling's variation is very small at the present field value. The same constraint applies to $\kappa(\phi)$ here, and it is the first of the requirements of Section 2.5: every excitation must respond to the medium in exactly the same proportion.

Earlier versions placed this framework in the second family. With the position adopted in this version (the preface and Section 1), its account of what is fundamental moves to the first: the medium is primary and geometry is its effective description. Its working equation remains that of the second family, Equation (1), with the medium's stresses on the right-hand side.

10.2 What is inherited and what is not

Inherited: the identification of the vacuum as a scalar condensate; the proposal that the dark sector is one medium rather than two substances; the use of bulk and shear viscosity to generate an effective pressure; the appearance of a coherence length in the galactic regime.

Inherited also, and more directly than the preceding list suggests: the generation of galactic structure by vorticity in a vacuum superfluid. Huang and collaborators [1] obtain vortices from galactic rotation

within a nonlinear Klein–Gordon description in which galaxies act as external sources. Section 6 reaches for the same phenomenon through different machinery — a dissipative stress tensor $\Pi_{\mu\nu}$ entering the field equation (1) rather than a source term in a wave equation. Whether these are distinct physical mechanisms or one mechanism in two formalisms is not settled here, and a credible version of this framework must say which.

The premise that ordinary matter is itself an excitation of the vacuum medium is also not new, and it is worth being precise about who holds it. Huang [1] and Berezhiani and Khoury [2] are two-species: the condensate constitutes the dark sector while baryonic matter is a separate species coupled to it. Superfluid vacuum theory is not. Its stated paradigm is a background quantum liquid of which "elementary particles", in its own terms, are excitations — one substance, as here.

What remains distinctive is narrower than earlier versions claimed. Superfluid vacuum theory obtains a single substance by making geometry emergent from the condensate, and this framework now takes the same position on what is fundamental, so that is no longer a point of difference. Two elements remain. The first is the route by which geometry is recovered: rather than deriving an acoustic metric from the condensate and then seeking Einstein's equations, the framework keeps Einstein's equation as the effective description and proposes a refractive account of why it holds (Section 2.5), whose decisive test is that the medium's response slows time and stretches space in equal measure for every excitation. Whether that route is genuinely different from the emergent programme's, or the same one approached from the other end, must be settled. The second is the mechanism of the partition, the covariant criterion of Section 7: the dimensionless quantity $\mathscr{Q} = \xi^2 h^{\mu\nu} k_\mu k_\nu$ of (16), with a smooth weighting $f(\mathscr{Q})$ splitting the mode integral between $T_{\mu\nu}^{\text{matter}}$ and $T_{\mu\nu}^{\phi}$. The consequence developed in Section 9 — that no discontinuous exchange term between the two is required, because both are populations of one field — follows from that construction. It removes an obstruction to the conservation conditions; it does not by itself establish them, which remains outstanding.

10.3 The electron-scale coincidence

The numerical agreement of Section 7 has been approached from the opposite direction. Working from a logotropic equation of state, Chavanis [18] recalls an expression for the cosmological constant in terms of the electron mass and the fundamental constants,

$$\Lambda = \frac{G^2 m_e^6}{\alpha^6 \hbar^4} \approx 1.36 \times 10^{-52}\,\mathrm{m}^{-2}, \tag{26}$$

a relation of Eddington type. There the electron mass fixes Λ; here the electron mass is the readout of $\xi(\phi_0)$. Both routes tie the electron scale to a vacuum-scale quantity, from opposite ends.

Two cautions attach to this. The relation is *recalled* rather than derived in that work, so the primary source lies earlier in the logotropic literature. And independent arrival at a coincidence is weak evidence that it is not accidental; it is not this framework's discovery, and the open question of Section 7.7, item (6), remains untouched by it.

10.4 Known deficiencies relative to current practice

Points where this document is behind the literature it should be engaging, or exposed to measurements it has not yet addressed.

(1) **First-order dissipative hydrodynamics.** The constitutive form of Section 2.3 is of Eckart–Landau type, which is known to be acausal and generically unstable in relativistic settings [19]. Current viscous cosmology uses causal second-order formulations of Israel–Stewart type [20]. A viable version of this framework must do the same.

(2) **Silence on vertical dynamics.** Section 6 states only a radial relation. Milky Way kinematics constrain the two components separately, and they do not want the same thing: the baryonic distribution supplies too little gravity to account for the rotation curve but enough to account

for the vertical velocity dispersions near the Sun, so substantial enhancement is required radially and essentially none vertically [10, 11]. Superfluid dark matter fails this test because its phonon-mediated force is approximately parallel to the baryonic gravitational field and therefore enhances both components by the same factor [11]. Cold dark matter passes because its vertical acceleration is dominated by baryons, with the halo subdominant. The resulting criterion applies to any model: the enhancement of vertical acceleration must be suppressed relative to the radial enhancement. Whether this framework is exposed depends on whether $v_{\phi,\mathrm{eff}}$ tracks the baryonic acceleration field — in which case it inherits the parallelism problem — or constitutes an independent source. Section 6 does not say which.

(3) **The status of $c(\phi)$.** A variable effective propagation speed is natural in the emergent-gravity family, where the metric is acoustic. It sits awkwardly alongside a fundamental $G_{\mu\nu}$. The framework must either justify $c(\phi)$ as a refractive effect within fixed geometry, or acknowledge that taking it as a genuine causal speed moves the proposal toward the emergent family and incurs that family's obligations. The refractive reading is now the framework's account of gravity (Section 2.5), which places it with the emergent family on what is fundamental and incurs that family's chief obligation: to derive the effective geometry rather than assume it.

(4) **The vacuum is already measured at the electron scale.** In quantum electrodynamics, vacuum polarization screens every charge out to the electron's reduced Compton wavelength — the same 386 fm as ξ_0 — and contributes measurably to the Lamb shift and to the electron's anomalous magnetic moment, in agreement with theory to many significant figures. The vacuum also behaves as a nonlinear optical medium. Light-by-light scattering, forbidden in classical electrodynamics, has been observed in ultra-peripheral heavy-ion collisions [21], and the strong linear polarization of optical light from the neutron star RX J1856.5−3754 is evidence for vacuum birefringence in a strong magnetic field [22]. The Casimir effect is measured. If ξ_0 is a property of the Ocean, these are the Ocean's responses at and near its healing length, and the framework must reproduce them quantitatively. Until it does, the electron-scale coincidence of Section 7 is matched by an obligation of the same order.

10.5 The structural question

If the vacuum is a medium possessing density, pressure and a rest frame, it is reasonable to ask why it requires a background spacetime to occupy. The emergent-gravity family answers that it does not, and that geometry is the medium's own collective behaviour. Earlier versions of this framework posited a medium and a geometry as distinct structures coupled through $\kappa(\phi)$. This version answers the question in favour of the medium. There is no background spacetime for the Ocean to occupy; the geometry of Equation (1) is how the medium's state is experienced by excitations of the medium, and $\kappa(\phi)$ expresses how strongly that state responds to concentrated energy. The position is now stated. It has not yet been derived, and deriving the effective geometry from the dynamics of the medium is the task it creates (Section 11).

11 Research Program and Testable Predictions

- Derive the field equations from a covariant action rather than introducing stress terms phenomenologically.
- Derive the effective equation of state $P(\phi)$ and determine whether a stable oscillatory cosmological solution exists.
- Solve the weak-field, stationary, rotating-galaxy limit and compare predicted rotation curves with baryonic distributions.

- Determine whether the same parameter set fits multiple galaxies without an independently chosen halo profile.
- Calculate the weak-field and post-Newtonian limits to verify recovery of Solar-System observations.
- Determine the propagation speed of scalar, gravitational, and electromagnetic perturbations, and identify whether $c(\phi)$ is a true causal speed or an effective refractive velocity.
- Test cosmological expansion, structure growth, gravitational lensing, and background-radiation constraints.
- **Derive the gradient-stiffness coefficient** $\mathscr{G}$, and with it the healing length $\xi(\phi)$ of (15), from the scalar Lagrangian.
- **Confront the predicted drift in** ξ — and hence in the effective electron mass — with quasar absorption spectra and Oklo constraints.
- **Resolve or explain the hierarchy** between the mass gap $\Omega_{\rm gap}$ required by galactic dynamics and the stiffness scale implied by $\xi_0 \approx 386\,\mathrm{fm}$. The candidate route of Section 7.7, item (5), divides this into three tasks.
  - *Protection.* Reformulate the scalar sector with ϕ as the phase of a complex field (5), re-express P, κ, ξ, η, ζ and the quartic potential of Section 9.1 in terms of the modulus or of X, and determine whether the shift symmetry survives the matter coupling and the dissipative sector. If it does not, compute the gap that results.
  - *Size.* Determine what fixes the extent of a galactic solution, given that the medium supplies no kiloparsec scale of its own: which features of the full state of the disturbance set it — energy, organised and random flow, exchange with the surrounding medium — and where their values originate.
  - *Rotation curves without per-galaxy freedom.* Derive $v_{\phi,\mathrm{eff}}(r)$ from the baryonic distribution alone, with no per-galaxy freedom, and compare with the observed radial acceleration relation [9, 8], including its universal acceleration scale $g_\dagger \approx 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$, which the framework must then account for, and extend the derivation to pressure-supported galaxies, which follow the same relation [8]. This must be done jointly with the vertical-acceleration item below, since a contribution fixed by the baryons is the kind most exposed to it.
- **Determine the sign of** λ from the chosen $L(\phi, X)$ and verify the defocusing condition of Section 9.1. A framework whose potential turns out to be focusing about the present vacuum has no mechanism arresting concentration and is not viable as written. Map the states in which stiffness is lost, determine what the medium does there, and confront the result with domain-wall, microwave-background and relic-abundance bounds.
- **Distinguish the galactic mechanism from its precedent.** Determine whether the rotational support of Section 6, generated by $\Pi_{\mu\nu}$, is physically distinct from the vortex formation obtained by Huang and collaborators [1] from a nonlinear Klein–Gordon equation with galaxies as external sources, or whether the two are the same mechanism expressed in different formalisms.
- **Establish the direction of the vacuum contribution.** Determine whether the acceleration produced by $\Pi_{\mu\nu}$ is parallel to the baryonic gravitational field, as a scalar-mediated MOND-like force is, or whether its anisotropic shear components yield a contribution with an independent orientation. This is the question on which the galactic sector turns: a parallel contribution enhances radial and vertical accelerations by a common factor and is excluded by local Milky Way data [10, 11], whereas an independent, quasi-spherical contribution is not. Then compute the predicted vertical acceleration explicitly and confront it, together with the rotation curve of Section 6, with stellar kinematics.

- **Reformulate the dissipative sector causally.** Replace the first-order constitutive relation of Section 2.3 with a second-order Israel–Stewart or BDNK form, and verify that the resulting system is causal and stable in the weak-field limit.
- **Settle the status of** $c(\phi)$ as either a refractive effect within fixed geometry or a genuine causal speed, and state the consequences for the framework's relation to emergent-gravity formulations (Section 10).
- **Establish whether matter coupling constitutes a forcing term.** The known finite-time blowup constructions for viscous media require an external force injecting momentum; the unforced problem remains open. In this framework the candidate forcing is the matter sector acting on the vacuum through $\kappa(\phi)T^{\text{matter}}_{\mu\nu}$ in (1). Determine whether that coupling can drive the medium to finite-time concentration, or whether the healing-length cutoff bounds it for any physically admissible matter distribution. This is the sharp form of the question raised in Section 5 as to whether a contraction phase terminates in a singularity or is arrested, and the answer turns on $\xi(\phi)$ and the sign of λ rather than on the viscosity coefficients.
- **Derive the viscosities from vortex dynamics.** Determine whether η and ζ follow from the creation, motion and reconnection of quantized vortices, as effective dissipation does in laboratory superfluids, removing two free functions from the theory.
- **Simulate a galactic disturbance.** Once the equation is fixed, solve the condensate wave equation (Gross–Pitaevskii or nonlinear Klein–Gordon type) for a galaxy-scale disturbance, including shear, wakes and vortex structure.
- **Reproduce the measured vacuum.** Compute the Ocean's contribution to vacuum polarization, and hence to the Lamb shift and the anomalous magnetic moment, together with light-by-light scattering, vacuum birefringence and the Casimir force, and compare with quantum electrodynamics and with measurement (deficiency (4) of Section 10).
- **Size and spin of the localized excitation.** Compute the probe response of the localized excitation and show it is pointlike to about 10^{-18} m; determine how $\xi(\phi)$ behaves at high concentration consistently with the collapse floor; and identify the structure that supplies spin $\frac{1}{2}$.
- **Motion through the surroundings.** Determine the flow regimes of a disturbance moving relative to the medium, including the superfluid critical speed. Test for vacuum contributions offset relative to the direction of motion; for residuals of the radial acceleration relation that track speed through the environment, a property not among those tested in [8]; and against colliding clusters.
- **Derive gravity from the medium.** Derive the effective geometry of Equation (1) from the dynamics of the medium, and show that the refractive account of Section 2.5 meets its three decisive requirements: every excitation responds in exactly the same proportion; time is slowed and space stretched in equal measure, giving the observed deflection of starlight; and gravitational waves travel at the speed of light. Then extend it to perihelion advance, frame dragging and gravitational radiation, or identify where it fails.
- **Black holes.** Determine whether collapse terminates near the healing length, whether the geometric and medium horizons coincide, and what horizon-scale imaging and ringdown measurements constrain (Section 8.1).
- **Hearing the medium.** If the gapless phonon of the candidate route couples to matter, it implies a weak long-range force or an energy-loss channel. Compare with torsion-balance and atom-interferometer bounds.

12 Summary of the Revised Formalism

The revised Ocean framework rests on a single premise: there is no empty space. The Ocean is one continuous medium without edges or outside, and everything in it, matter and light included, is a pattern of that medium. Gravity is proposed to be the medium's response to concentrated energy. The mathematical formulation keeps Einstein's equation as the effective description of that response, with a scalar-field stress sector, a viscous constitutive sector, a variable pressure contribution and a field-dependent effective coupling as the medium's contributions to it.

The framework extends that idea to the matter sector itself. If the vacuum is a medium, it has a characteristic length over which it can heal; excitations shorter than that length are localized patterns, and excitations longer than it are motions of the medium. The distinction between matter and vacuum then becomes a statement about the medium's own excitation spectrum rather than an imposed energy cut.

The central hypothesis is that phenomena conventionally divided into dark energy, dark matter, and localized matter may have a common origin in different dynamical regimes of one vacuum field. Establishing that claim requires deriving the relevant limits and demonstrating quantitative agreement with observations.

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Reference basis. Revised from an earlier formalism supplied by the author. That document defines the Ocean metric, the vacuum viscosity tensor, the scalar pressure term, the effective coupling $\kappa(\phi)$, the VSL interpretation, and the proposed 0.511 MeV matter threshold. The matter-threshold section of that document is superseded by Section 7 below.